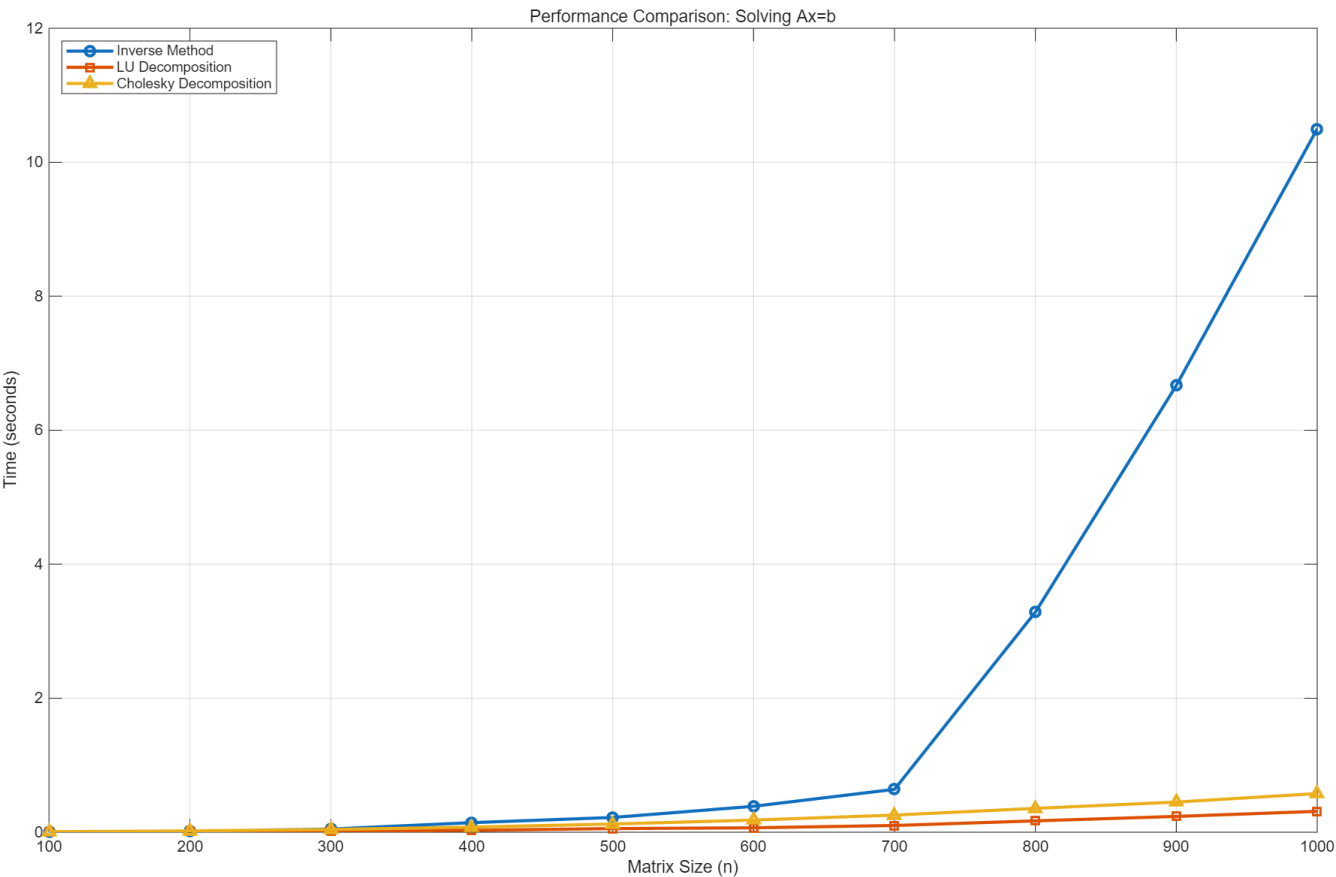
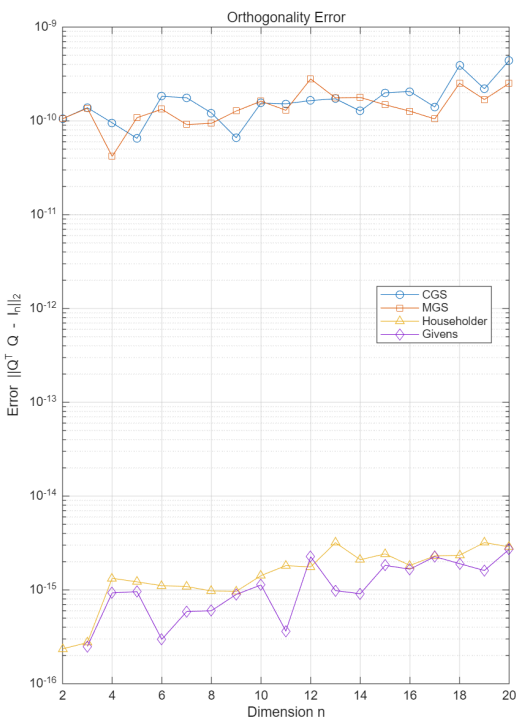
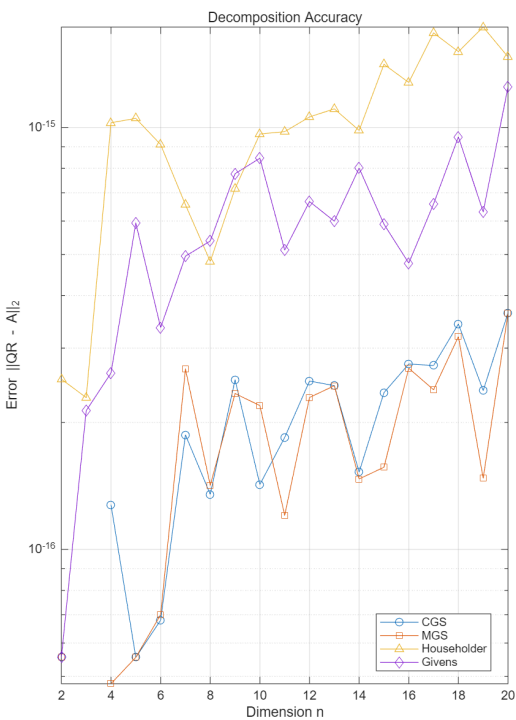


Homework 1

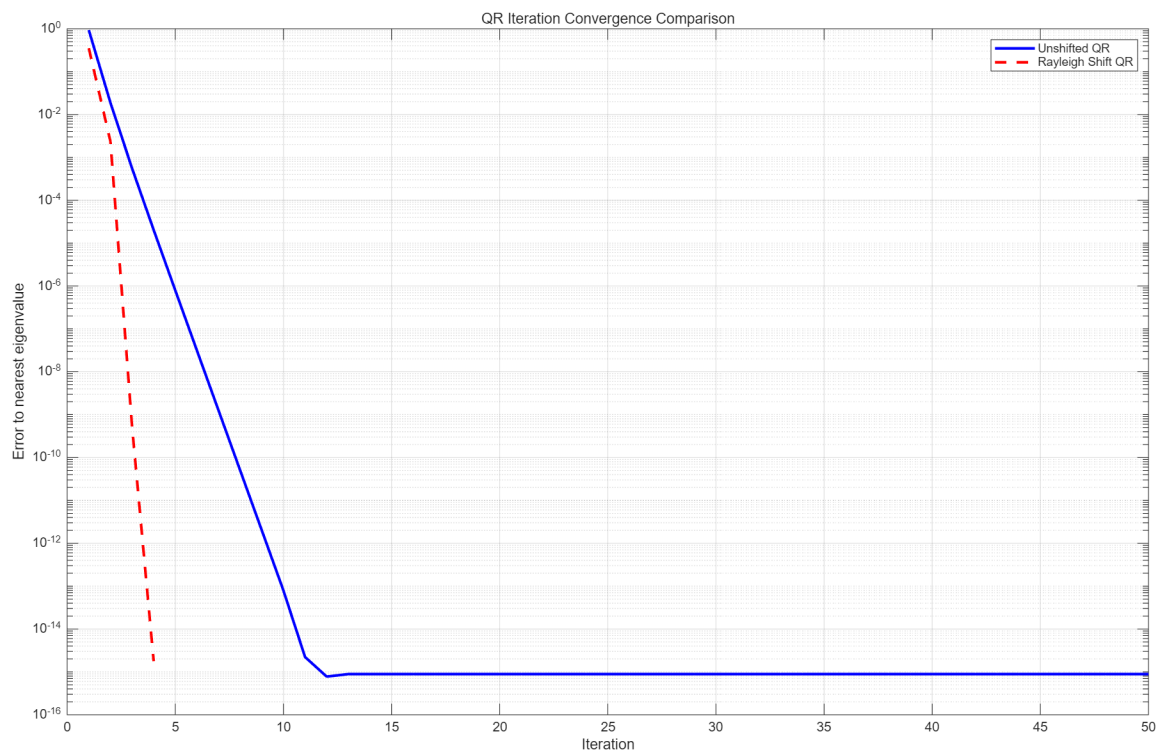


Homework 2

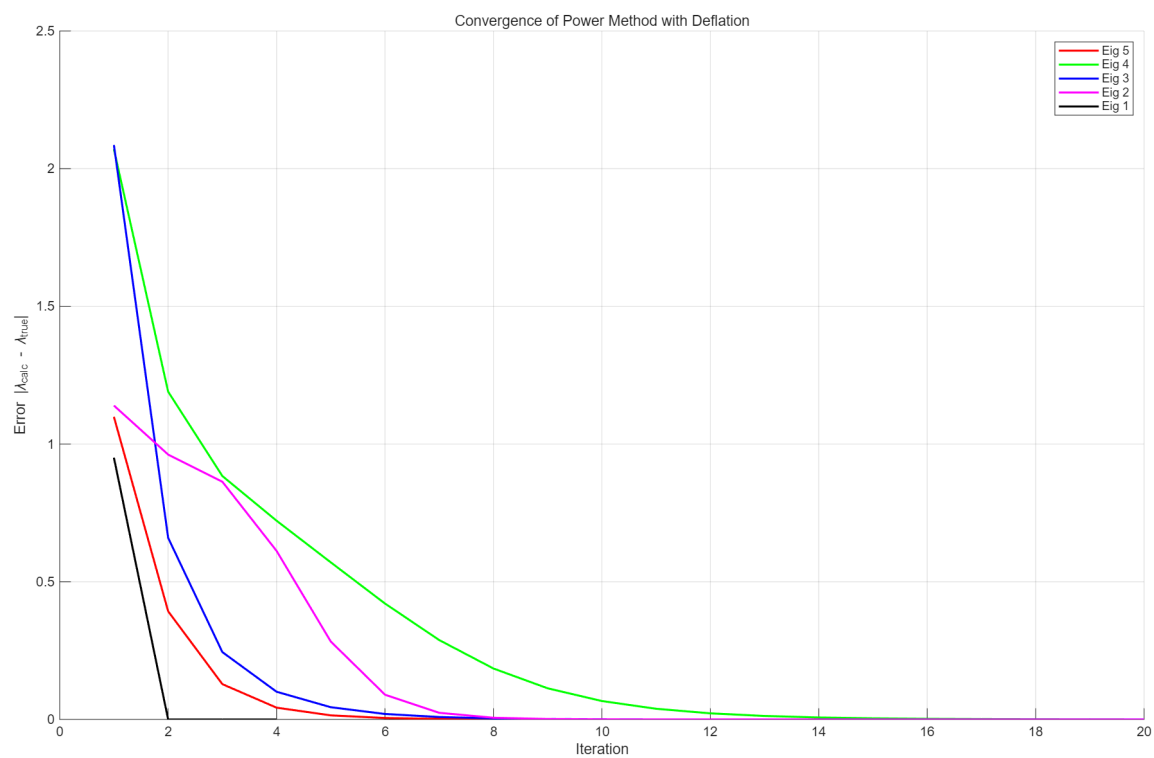


Homework 3

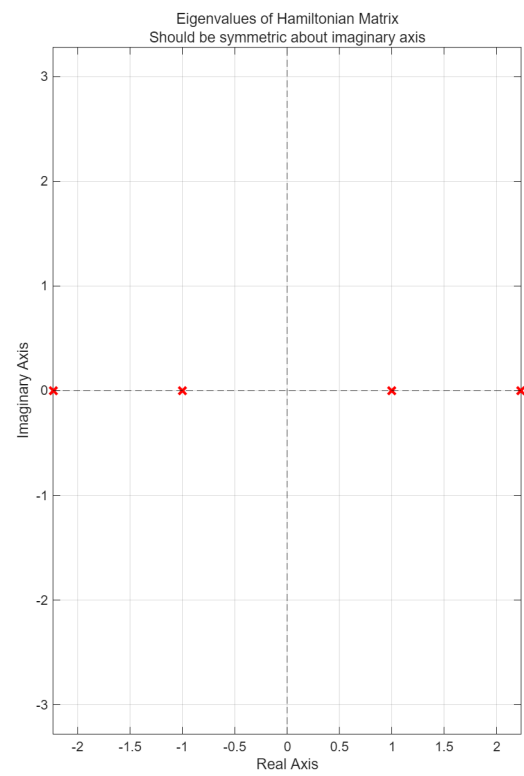
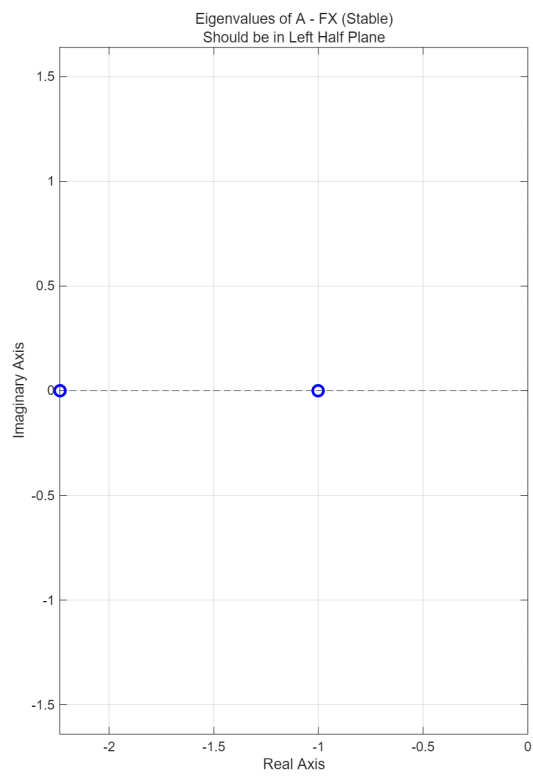
Problem 1



Problem 2



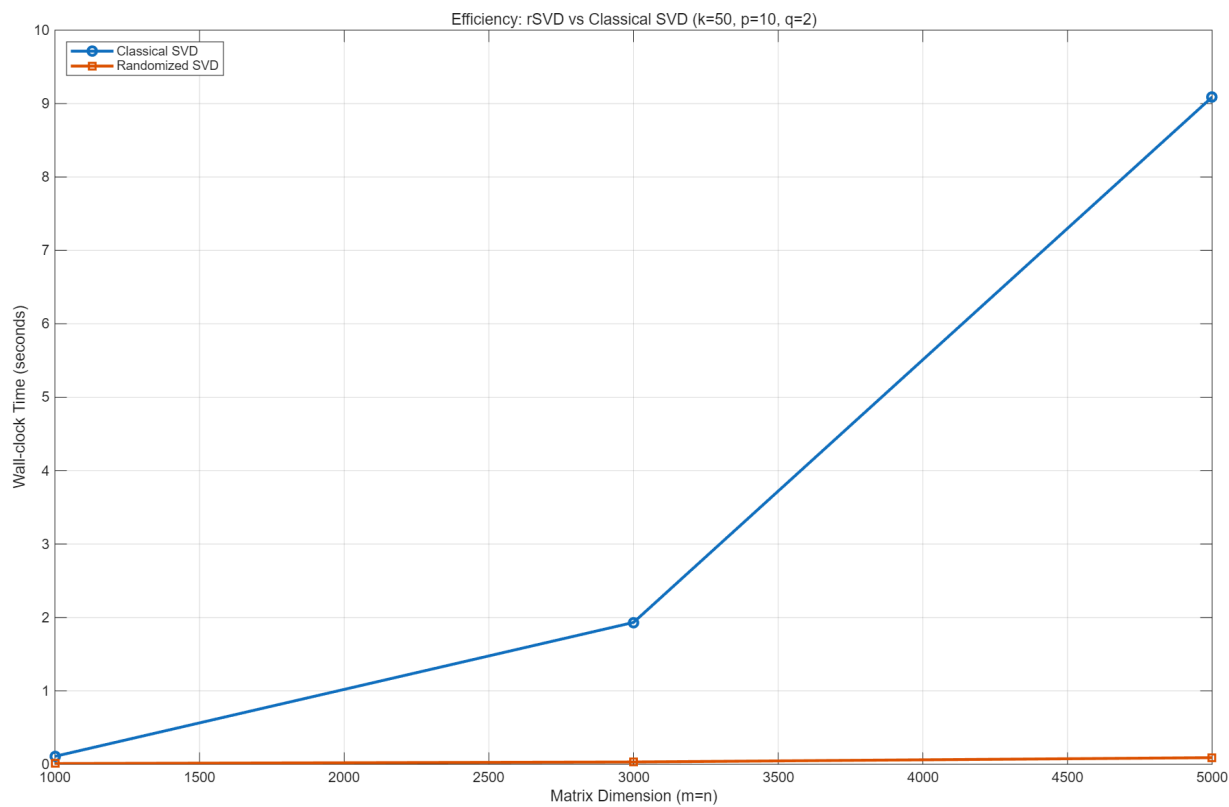
Problem 3



Homework 4

Problem 1

Matrix Dimensions	Classical SVD Time (s)	Randomized SVD Time (s)	Speedup Ratio
1000 * 1000	0.1006	0.0092	10.95
3000 * 3000	1.9701	0.0273	72.11
5000 * 5000	9.1197	0.0735	124.10



Problem 2

Oversampling (p)	Power Iterations (q)	Relative Frobenius Error	PSNR (dB)
0	0	1.3614e-01	22.90
0	1	9.6462e-02	25.90
0	2	9.1724e-02	26.33
0	3	9.0476e-02	26.45
5	0	1.2930e-01	23.35
5	1	9.2562e-02	26.25
5	2	9.0185e-02	26.48
5	3	8.9344e-02	26.56
10	0	1.1796e-01	24.15
10	1	9.0736e-02	26.43
10	2	8.9400e-02	26.56
10	3	8.9307e-02	26.56
20	0	1.1193e-01	24.60
20	1	8.9803e-02	26.52
20	2	8.9200e-02	26.58
20	3	8.9190e-02	26.58

